MIL-T-51289B(EA)
22 May 1985
SUPERSEDING
MIL-T-51289A(MU)
12 August 1970

MILITARY SPECIFICATION

TEST SET, FLAME THROWER - RIOT CONTROL AGENT DISPERSER, HYDROSTATIC - VOLUMETRIC, 6000 PSI, M5

This specification is approved for use by U.S. Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a test set for hydrostatic and volumetric testing portable flame throwers and agent dispersers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation forma part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. ArmY Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

FSC 1040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-c-843 - Cushioning Material, Cellulosic.
PPP-F-320 - Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291 - Paperboard, Wrapping and Cushioning.
PPP-T-60 - Tape: Packaging, Waterproof.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Packaging/Paper (For Carton Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-B-2427? - Box, Ammunition Packing: Wood, Nailed.

STANDARDS

MILITARY

MIL-STD-105 - sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Ammunition Lot Numbering.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

TDPL 81-6-392 - Test Set, Flame Thrower - Riot Control Agent Disperser, Hydrostatic - Volumetric, 6000 psi, M5.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 81-6-392 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawing.

3.1.2 Components. All components of the kit shall conform to the specifications and drawings listed on TDPL 81-6-392 and subsidiary drawings.

3.2 Pump leakage. The pump shall not leak when pressurized to 3750 + 50 or 0 pounds per square inch gage (psig) for a period of 30 + 1 or -0 minutes when tested as specified in 4.4.5.

3.3 Kit. The components shall be assembled into a kit as specified in section 5.

3.4 First article. Prior to the start of regular production, a first article lot of test set components shall be produced for examination and tests in accordance with this specification.

3.5 Workmanship. The components of the kit shall be free from contamination (foreign matter, oil or grease) and damage such as dents, cracks or deformation that could affect their use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(C) packaging inspection (see 4.4).

4.3 First article inspection.

4.3.1 Lot. A first article lot of five each of all components, subassemblies, assemblies and packaging (section 5) shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.l For examination and tests. The sample components subassemblies, assemblies and packaging shall be examined and tested for all specification and drawing requirements.

4.3.2.2 Acceptance/rejection criteria. The first article sample shall comply with the examination and tests specified in 4.3.2.1 to be acceptable. The acceptance number is zero. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot of test sets shall be identified and controlled in accordance with MIL-STD-1168.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For leakage test. There is no sampling. Each pump in the lot shall be tested.

4.4.3 Inspection procedure.

4.4.3.1 For examination. The test set packaging shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For leakage test. Each pump shall be tested in accordance with 4.4.5. Any pump failing the test shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:56Z
Check the source to verify that this is the current version before use.

4.4.4 Classification of defects.

(a) Packaging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Components improperly cleaned and dried "
102	Components missing or incorrectly packaged
103	Containers incorrect, broken or crushed
104	Container closure incorrect
105	Cushioning missing or incorrect
106	Preservatives missing or incorrect
107	Marking missing, incorrect or illegible
108	Strapping loose or missing

4.4.5 Pump leakage test. The pump shall be subjected to specified internal hydrostatic pressure for a period of 30 + 1 or - 0 minutes and observed for leakage (see 6.4).

5. PACKAGING

5.1 Cleaning and drying. The hydraulic pump assembly and other kit components shall be cleaned and dried in accordance with MIL-P-116, using method C-1 and any of the drying methods listed therein.

5.2 Preservation and packaging. Level shall be as specified (see 6.2).

5.2.1 Level A. Unless otherwise specified, preservatives and the application thereof and packaging methods shall conform to MIL-P-116.

5.2.1.1 Hydraulic pump assembly. The pump assembly, less gage, bleeder valve, pipe elbow, nipple, pump lever handle and pressure relief valve lever shall be preserved by lubricating the hydraulic pump in accordance with the manufacturer's instructions and coating the interior of the hydraulic storage tank with type P-9 preservative. Fill the tank with preservative and then drain and coat the fill and drain plugs with the same type preservative and replace plugs and tighten. Other threaded openings in the pump shall be coated with preservative and the openings closed with plastic caps or plugs or with tape Conforming to type optional, class 1 of PPP-T-60. All unpainted exterior metal surfaces of the pump shall be coated with P-9 preservative.

5.2.1.2 Valve and hose assembly. The valve attached to the hose and the hose fittings shall have both internal and external surfaces coated with P-9 preservative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

5.2.1.3 Miscellaneous metal components, fittings, pressure relief control lever and pump lever handle. Unpainted external metal surfaces shall be coated with P-9 preservative.

5.2.1.4 Hydraulic pump and components. The pump, preserved in accordance with 5.2.1.1, shall be wrapped in barrier material conforming to type I, grade A of MIL-B-121 and then placed in a snug fitting fiberboard box conforming to type CF, class weather-resistant.
ney type fiberboard cell structure shall extend between the bottom flange of the tank and the top of the box to support the pump and tank assembly. The pump lever shall be placed in the down position. The pump lever and pressure release control lever shall have preservative areas covered by barrier material cited above and then wrapped in paperboard cushioning material conforming to type 111 of PPP-P-291, secured in place with tape conforming to PPP.T-76. The unattached fittings, preserved in accordance with 5.2.1.3, shall be wrapped in barrier material cited above, except the pipe nipple shall have the threaded ends only covered and secured with above cited tape. The valve, lever, and fittings shall be included in the fiberboard box with the pump and tank. Cushioning material conforming to type 11, class B of PPP-c-843 shall be used to restrict free movement of the contents. The box shall be closed and water-proofed in accordance with the appendix to PPP-B-636. The pump lever handle requires no packaging.

5.2.1.5 Valve and hose assembly. The valve and hose assembly, preserved in accordance with 5.2.1.2, shall have preservative-treated areas wrapped in barrier material, same as previously used. The hose shall then be compactly coiled, without kinking, and packaged in a snug-fitting fiberboard box conforming to type CF, class weather-resistant, variety SW, grade w6c of PPP-B-636. Cushioning material, same as previously used, shall be used to restrict free movement of the hose. The box shall be closed and waterproofed in accordance with the appendix to PPP-B-636.

5.2.1.6 Gage, bleeder valve assembly and “O” ring gaskets. Two “O” ring gaskets shall be sealed in a polyethylene bag, 0.004 inch minimum thickness. A paperboard stiffener shall be included in the bag to prevent deformation of the gaskets. The gage and bleeder valve assembled and the gaskets shall be packaged in accordance with method IIb. Preservative is not required. Cushioning material, same as previously used, shall be used to prevent damage and restrict free movement of the items. The outer box shall conform to type CF, class weather-resistant, variety SW, grade w6c of PPP-B-636.

5.2.1.7 Test stand assembly. The test stand assembly shall be packaged in a snug-fitting fiberboard box coforming to type CF, class weather-resistant, variety SW, grade W5c of PPP-B-636. Cushioning material, same as previously used, shall be used to prevent free movement of the test stand. The box shall be closed and waterproofed in accordance with the appendix to PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:56Z
Check the source to verify that this is the currentversion before use.

5.2.1.8 Technical publications. The technical publications shall be packaged together in a polyethylene bag, 0.006 inch minimum thickness. The bag shall be sealed keeping included air volume to a minimum.

5.2.2 Level C. Preservation shall be in accordance with 5.2.1.

5.2.2.1 Hydraulic pump. The hydraulic pump and components shall be packaged in accordance with 5.2.1.4, except the box shall conform to type CF, class domestic, variety SW, grade 275 of PPP-B-636 and the closed box need not be waterproofed,

5.2.2.2 Gage, bleeder valve assembly and "O" ring gaskets. The gaskets shall be packaged in accordance with 5.2.1.6 and packaged with the gage and bleeder valve assembled in a fiberboard box conforming to type SW, class domestic, variety SW, grade 175 of PPP-B-636. Cushioning material, same as previously used, shall be used to prevent damage and free movement of the items. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.2.3 Test stand assembly. The test stand assembly shall be packaged in accordance with 5.2.l.7, except the fiberboard box shall conform to type SW, class domestic, variety SW, grade 275 of PPP-B-636 and the closed box need not be waterproofed.

5.2.2.4 Technical publications. the technical publications shall be packaged in accordance with 5.2.1.8.

5.3 Packing. Level shall be as specified (see 6.2).

5.3.1 Level A. The test set and components, packaged in accordance with 5.2.1, shall be packed together in a cleated-plywood box conforming to overseas type, style A, grade A, type 3 (difficult) load, 100 to 200 pound load limit of PPP-B-601. Prior to packing items in the box, it shall meet the dryness requirements of MIL-B-2427. The box shall fit the contents snugly to keep cube displacement to a minimum, including variety SW, grade W5C, weather-resistant fiberboard filler pads conforming to PPP-F-320. Blocking or bracing shall be used as necessary to restrict free movement of the contents. The technical publications shall be positioned in box so as to be readily available when opened. The box shall be closed and strapped in accordance with the appendix to PPP-B-601. Strapping shall conform to type I or IV, class B of QQ-S-781.

5.3.2 Level B. The test set and components, packaged as specified, shall be packed in accordance with 5.3.1, except wood preservative is not required and strapping shall conform to class A.

5.3.3 Level C. The test set and components, packaged as specified, shall be packed in accordance with 5.3.1, except the plywood box shall conform to domestic class and wood preservative is not required and strapping shall conform to class C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

5.4 Marking. In conjunction with information furnished in the contract or order, marking shall be as specified herein:

5.4.1 Package. The outer wrap of each package shall be marked "Test Set, Flame Thrower - Riot Control Agent Disperser, Hydrostatic - Volumetric, 6000 PSI, M5, Component of." Below the above marking, list the contents of the package as identified on PL 81-6-392. Also indicate the level of packaging and date packaged. Marking shall be in accordance with MIL-STD-129. Transparent plastic bags require no marking, however contents shall be identified in accordance with specified requirements.

5.4.2 Packing. The box shall be marked in accordance with Drawing 81-6-418.

6. NOTES

6.1 Intended use. This set is used for testing fuel tanks, agent tanks, hoses and air cylinders of flame throwers and riot control agent dispersers.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.

6.3 Quality assurance provisions (QAPs).

QAP Number	Title
31-15-1297	Cap
81-1-2147	Body, Check Valve
81-1-2452	Stem
81-1-2453	Handle, Stem
81-1-4883	Plug Assembly, Filling, C23
81-6-254	Adapter, Testing Gage
81-6-380	Adapter, Testing Gage
81-6-381	Adapter, Tank
81-6-384	Adapter, Straight Pipe to Tube
81-6-393	Valve and Hose Assembly
81-6-394	Plug, Solid, Diffusion Pipe (M2A1-7)
81-6-395	Plug, Assembly
81-6-397	Adapter, Pressure Bottle, Hose.
81-6-399	Plug, Solid, Diffusion Pipe (M9-7)
81-6-402	Rack Assembly
81-6-403	Large Tube Assembly
81-6-404	Test Stand, Volumetric, Pressure, Assembly
81-6-406	Small Tube Assembly
81-6-407	Valve, Assembly
81-6-409	Scale Assembly
81-6-411	Reducer
81-6-421	Adapter

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

6.4 Caution. Due to the hazards inherent in testing with high pressure, contractors are advised that every precaution possible should be exercised for the safety and protection of personnel engaged in such testing.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify *changes* with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army)IA

Preparing activity:

Army-EA

Project No. 104O-Al10

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:56Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament, Munitions and Chemical Command
ATTN: AMSMC-TDC-S (A)
Aberdeen Proving Ground, MD 21010-5423

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:56Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-51289B(EA)	TEST SET, FLAME THROWER - RIOT CONTROL AGENT DISPERSER, HYDROSTATIC - VOLUMETRIC, 6000 PSI, M5

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark o*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 EDITION OF 1 OCT 78 IS OBSOLETE

Source: https://assist.dla.mil - Downloaded: 2026-04-03T20:56Z
Check the source to verify that this is the current version before use.

NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-T-51289B(EA)
NOTICE 1
28 September 1995

MILITARY SPECIFICATION

TEST SET, FLAME THROWER-RIOT CONTROL AGENT DISPERSER HYDROSTATIC - VOLUMETRIC, 6000 PSI, M5

This notice should be filed in front of MIL–T–51289B(EA) dated 22 May 1985.

MIL–T–51289B(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity:

Army - EA

Project No. 104O-A217

AMSC N/A

FSC 1040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:56Z
Check the source to verify that this is the current version before use.